

The Hamiltonian path passing through prescribed edges in a star graph with faulty edges *

Tsung-Yu Yu and Chun-Nan Hung

Department of Computer Science and Information Engineering

Da-Yet University, Changhua, Taiwan 51591, R.O.C

r9806027@mail.dyu.edu.tw, spring@mail.dyu.edu.tw

Abstract

Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . In [5], the authors showed that E_0 lies on a Hamiltonian path $P(u, v)$ of $S_n - F_e$ where $d(u, v)$ is odd, $|F_e| \leq n - 3$, $|E_0| \leq 2n - 7 - 2|F_e|$. In this paper we improve the previous result as follows: E_0 lies on a Hamiltonian path $P(u, v)$ of $S_n - F_e$ where $d(u, v)$ is odd, $F_e \leq n - 3$, $|E_0| \leq 2n - 6 - 2|F_e|$.

1 Introduction

The hypercube Q_n and star graph S_n are the most popular and efficient interconnection networks. They possess many excellent properties such as recursive structure, symmetry, small diameter, low degree, popular structure embedding, easy routing, and optimal connectivity and fault-tolerance. There are many literatures for fault tolerance of Hamiltonian properties of hypercube and star graphs.

The interconnection network is usually represented by an undirected graph where vertices represent processors and edges represent links. Let $G = (V_0 \cup V_1, E)$ be a *bipartite graph* where V_0 and V_1 are two disjoint vertex sets such that each edge of E consists of one vertex from each set. A *Hamiltonian path* is a spanning path of G . A *Hamiltonian cycle* is a cycle that visits every vertex exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle. Let F_e be a set of faulty edges and $|F_e|$ be the number of F_e . A graph $G = (V, E)$ is *k edges Hamiltonian* if $G - F_e$ is a Hamiltonian graph for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph

$G = (V_0 \cup V_1, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_0, y \in V_1$. A Hamiltonian-laceable graph $G = (V_0 \cup V_1, E)$ is *hyper-Hamiltonian laceable* if for any vertex $v \in V_i, i = 0, 1$, there is a Hamiltonian path of $G - \{v\}$ between any two vertices of V_{1-i} . A bipartite graph $G = (V_0 \cup V_1, E)$ is *k edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable for every $F_e \subset E$ and $|F_e| \leq k$. In [3], Li et al. proved that S_n is $(n - 3)$ edges Hamiltonian laceable and $(n - 4)$ edges hyper-Hamiltonian laceable with $n \geq 4$.

On the other hand, Dvořák investigated the problem about prescribed edges. Given a set of fault-free prescribed edges in a hypercube, there is a Hamiltonian cycle passing through all edges of this set in the hypercube. In [1], they showed that Q_n contains a Hamiltonian cycle passing through all edges of any set of prescribed edges E_0 if and only if the subgraph induced by E_0 consists of pairwise vertex-disjoint paths and $|E_0| \leq 2n - 3$. Wang et al. further presented the result that all edges of E_0 lie on a Hamiltonian cycle of $Q_n - F_e$ if E_0 is a linear forest (i.e., pairwise vertex-disjoint paths) and $|F_e| \leq n - 2, |E_0| \leq 2n - 3 - 2|F_e|$ in [4]. Let $d(x, y)$ be the distance of vertices x and y . The authors of [2] introduced that all edges of E_0 lie on a Hamiltonian path $P(x, y)$ of Q_n where E_0 is a linear forest with $|E_0| \leq 2n - 4$ and at most one vertex of x, y is incident with one edge of E_0 and $d(x, y)$ is odd.

Hung et al.[5] proved that there all edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n - 3, |E_0| \leq 2n - 5 - 2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n - 3, |E_0| \leq 2n - 7 - 2|F_e|$. In this paper, we improve the results of [5]. We prove that there is a Hamiltonian path $P(x, y)$ passing through all edges in E_0 of $S_n - F_e$ if $|F_e| \leq n - 3$ and $|E_0| \leq 2n - 6 - 2|F_e|$ for $n \geq 4$ where E_0 is a linear forest.

*The corresponding author is Tsung-Yu Yu and his email is r9806027@mail.dyu.edu.tw

2 Preliminaries

In this section, we will introduce some definition, notations and properties about star graphs which are useful for our proofs.

Definition 1 The n -dimensional star graph is denoted by S_n . The vertex set $V = \{v | v \text{ is a permutation of } 1, 2, \dots, n\}$ and edge set $E = \{(v_1 v_2 \dots v_{i-1} v_i \dots v_n, v_i v_2 \dots v_{i-1} v_1 v_{i+1} \dots v_n) | v_1 \dots v_n \in V \text{ and } 2 \leq i \leq n\}$

A star graph S_n is an edge symmetric and node symmetric graph. By definition, S_n contains $n!$ vertices and $\frac{n!(n-1)}{2}$ edges. The graph S_n can be decomposed to n copies of $(n-1)$ -dimensional star graphs. Give $2 \leq k \leq n$, we can use S_n^i to denote the subgraph of S_n induced by the vertices whose n th bit is i . We further denote the induced subgraph $S_n^{i_1} \cup S_n^{i_2} \dots \cup S_n^{i_k}$ as $S_n^{i_1, i_2, \dots, i_k}$ for $1 \leq i_1, i_2, \dots, i_k \leq n, 1 \leq k \leq n$. Let $\langle n \rangle = \{1, 2, \dots, n\}$ and I_k be the subset of $\langle n \rangle$ with $|I_k| = k$. Let u be a vertex of S_n^i for some $1 \leq i \leq n$. Let $[u]_k$ be the k th digit u . Let $\phi(u)$ be a vertex of S_n^j for $(u, \phi(u)) \in E(S_n)$ and $i \neq j$.

Lemma 1 If x and $y \in S_n$ and $d(x, y) \leq 2$, then $[x]_1 \neq [y]_1$.

Li et al. proved the follow lemma in [3].

Lemma 2 The graph S_n is $(n-3)$ -edge Hamiltonian laceable and $(n-4)$ -edge hyper-Hamiltonian laceable with $n \geq 4$.

Lemma 3 (See[5].) Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq 1, |E_0| \leq 3 - 2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ of S_n where $d(u, v)$ is odd, $|E_0| = 1$ for $n \geq 4$.

Lemma 4 (See[5].) Let $I = \{2, 3, \dots, n\}$. Let s and t be arbitrary vertices of $S_n^I = (V_0 \cup V_1, E)$ and $e \in E$ be a prescribed edge for $s \in V_0, t \in V_1, e \neq (s, t)$ and $n \geq 5$. There exists a Hamiltonian path $P(s, t)$ passing through the edge e of S_n^I .

Lemma 5 (See[5].) Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian cycle of $S_n - F_e$, if $|F_e| \leq n-3, |E_0| \leq 2n-5-2|F_e|$ and lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n-3, |E_0| \leq 2n-7-2|F_e|$.

3 Hamiltonian paths passing through prescribed edges

In this section, we will show that all edges of E_0 lie on a Hamiltonian path $P(x, y)$ of $S_n - F_e$ if $|F_e| \leq n-3$ and $|E_0| \leq 2n-6-2|F_e|$ for $n \geq 4$ where E_0 is a linear forest.

Lemma 6 Let $I = 2, 3, \dots, n$. Let s_1, s_2 and t_1, t_2 be vertices of $S_n^I = (V_0 \cup V_1, E)$ for $s_1, s_2 \in V_0, t_1, t_2 \in V_1, [s_1]_1, [t_2]_1 = 1, n \geq 5$. There exist two spanning disjoint paths $P(s_1, t_1)$ and $P(s_2, t_2)$ of S_n^I .

Proof. Let $U = \{s_1, s_2, t_1, t_2\}$ and $U^i = U \cap V(S_n^i)$ for $2 \leq i \leq n$. We will prove this lemma with the following two lemmas.

Case 1: $|U^i| = 4$ for some $2 \leq i \leq n$.

Without loss of generality, we can assume that $i = 2$. Applying Lemma 2, we can construct a Hamiltonian path between s_1 and t_1 of S_n^2 . Without loss of generality, we can denote this Hamiltonian path as $\langle s_1 \xrightarrow{P(s_1, x)} x, s_2 \xrightarrow{P(s_2, t_2)} t_2, y \xrightarrow{P(y, t_1)} t_1 \rangle$. Suppose that $\phi(x), \phi(y) \in S_n^j$ for some $3 \leq j \leq n$. Without loss of generality, we can assume that $j = n$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(x) \xrightarrow{P(\phi(x), a)} a, b \xrightarrow{P(b, \phi(y))} \phi(y), y \xrightarrow{P(y, t_1)} t_1 \rangle$ of S_n^n with $\phi(b) \in S_n^{k_1}$ and $\phi(a) \in S_n^{k_2}$ for $3 \leq k_1, k_2 \leq n-1$. We can also construct the Hamiltonian path $P(\phi(b), \phi(a))$ of $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, x)} x, \phi(x) \xrightarrow{P(\phi(x), a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(b))} \phi(b), b \xrightarrow{P(b, \phi(y))} \phi(y), y \xrightarrow{P(y, t_1)} t_1 \rangle$ and $P(s_2, t_2)$ are two spanning paths of S_n^I , as illustrated in Figure 1.

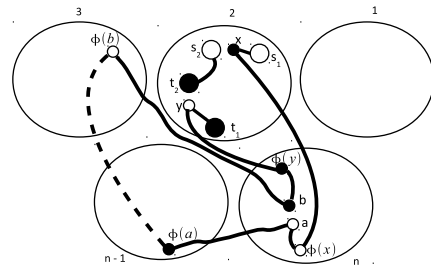


Figure 1: The illustration of case 1 of Lemma 6.

Suppose that $\phi(x) \in S_n^i, \phi(y) \in S_n^j, 3 \leq i \neq j \leq n$. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ of

$S_n^{3,4,\dots,n}$. Thus, $\langle s_1 \xrightarrow{P(s_1,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y) \xrightarrow{P(\phi(y),t_1)} t_1 \rangle$ and $P(s_2, t_2)$ are two spanning paths of S_n^I .

Case 2: $|U^i| = 3$ and $|U^j| = 1$ for some $2 \leq i \neq j \leq n$.

Without loss of generality, we can assume that $i = 2, j = n$ and $s_1, s_2 \in S_n^2$.

Case 2.1 $t_2 \in S_n^2, t_1 \in S_n^n$.

Applying Lemma 2, we can construct a Hamiltonian path $\langle s_1 \xrightarrow{P(s_1,x)} x, s_2 \xrightarrow{P(s_2,t_2)} t_2 \rangle$ of S_n^2 with $\phi(x) \notin S_n^1$.

Suppose that $\phi(x) \in S_n^n$. Applying Lemma 1, we can obtain that $(\phi(x), y)$ and (y, w) are two edges of S_n^n with $\phi(y) \in S_n^3$ and $\phi(w) \in S_n^1$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(x), y, z \xrightarrow{P(z,t_1)} t_1 \rangle$ of $S_n^n - \{(y, w)\}$ with $[z]_1 \notin \{1, 2, n\}$. We can construct the Hamiltonian path $P(\phi(y), \phi(z))$ of $S_n^{3,\dots,n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1,x)} x, \phi(x), y, \phi(y), \xrightarrow{P(\phi(y),\phi(z))} \phi(z), z, \xrightarrow{P(z,t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2,t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 2.

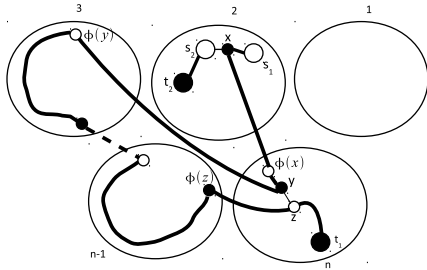


Figure 2: The illustration of case 2.1 of Lemma 6.

Suppose that $\phi(x) \notin S_n^n$. Applying Lemma 2, we can construct the Hamiltonian path $P(\phi(x), t_1)$ of $S_n^{3,\dots,n-1,n}$. Thus, $\langle s_1 \xrightarrow{P(s_1,x)} x, \phi(x), \xrightarrow{P(\phi(x),t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2,t_2)} t_2 \rangle$ are two spanning paths of S_n^I .

Case 2.2 $t_1 \in S_n^2, t_2 \in S_n^n$.

Applying Lemma 2, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2,x)} x, s_1 \xrightarrow{P(s_1,t_1)} t_1 \rangle$ of S_n^2 with $x \notin S_n^n$. Applying Lemma 2, we can construct the Hamiltonian path $P(\phi(x), t_2)$ of

$S_n^{3,\dots,n}$. Thus, $P(s_1, t_1)$ and $\langle s_2 \xrightarrow{P(s_2,x)} x, \phi(x) \xrightarrow{P(\phi(x),t_2)} t_2 \rangle$ are two spanning paths of S_n^I .

Case 3: $|U^i| = 2$ and $|U^j| = 2$ for some $2 \leq i \neq j \leq n$.

Without loss of generality, we can assume that $i = 2, j = n$ and $t_2 \in S_n^2$.

Case 3.1 $s_1 \in S_n^2$.

Applying Lemma 2, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2,u)} u, v \xrightarrow{P(v,t_1)} t_1 \rangle$ of S_n^n for $\phi(u), \phi(v) \notin (S_n^1 \cup S_n^2)$. Without loss of generality, we can assume that $\phi(u) \in S_n^{n-1}$ and $\phi(v) \in S_n^3$. Applying Lemma 1, we can obtain that (s_1, y) is an edge in S_n^2 with $\phi(y) \in S_n^3$. Applying Lemma 2, we can construct a Hamiltonian path $\langle s_1, y, x \xrightarrow{P(x,t_2)} t_2 \rangle$ of S_n^2 for $\phi(x) \in S_n^4$. We can also construct the Hamiltonian path $P(\phi(u), \phi(x))$ of $S_n^{4,\dots,n-1}$. Thus, $\langle s_1, y, \phi(y) \xrightarrow{P(\phi(y),\phi(v))} \phi(v), v \xrightarrow{P(v,t_1)} t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2,u)} u, \phi(u) \xrightarrow{P(\phi(u),\phi(x))} \phi(x), x \xrightarrow{P(x,t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 3.

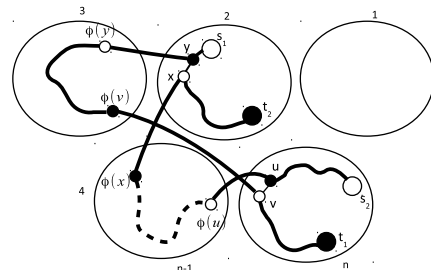


Figure 3: The illustration of case 3.1 of Lemma 6.

Case 3.2 $s_2 \in S_n^2$.

Applying Lemma 2, we can construct Hamiltonian path $\langle s_1 \xrightarrow{P(s_1,x)} x, y \xrightarrow{P(y,t_1)} t_1 \rangle$ of S_n^n with $[x]_1, [y]_1 \notin \{1, 2\}$. We can also construct the Hamiltonian path $P(s_2, t_2)$ of S_n^2 . We can construct the Hamiltonian path $P(\phi(x), \phi(y))$ in $S_n^{3,\dots,n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,t_1)} t_1 \rangle$ and $P(s_2, t_2)$ are two spanning paths of S_n^I , as illustrated in Figure 4.

Case 3.3 $t_1 \in S_n^2$.

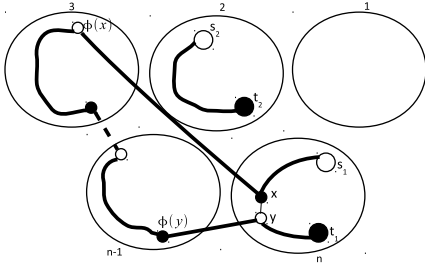


Figure 4: The illustration of case 3.2 of Lemma 6.

Let x be a neighbor of t_1 in S_n^2 with $\phi(x) \in S_n^n$. Applying Lemma 2, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, \phi(y))} \phi(y), s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x) \rangle$ of S_n^n for $y \notin S_n^{1,2}$. Without loss of generality, we can assume that $y \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle x, t_1, z \xrightarrow{P(z, t_2)} t_2 \rangle$ of S_n^2 for $\phi(z) \in S_n^3$. We can also construct the Hamiltonian path $P(y, \phi(z))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x, t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(y))} \phi(y), y \xrightarrow{P(y, \phi(z))} \phi(z), z \xrightarrow{P(z, t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 5.

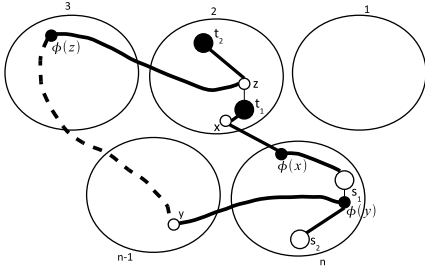


Figure 5: The illustration of case 3.3 of Lemma 6.

Case 4: $|U^i| = 2$, $|U^j| = 1$ and $|U^k| = 1$ for some different integers $2 \leq i, j, k \leq n$. Without loss of generality, we can assume that $i = 2$, $j = 3$ and $k = n$.

Case 4.1 $s_1, s_2 \in S_n^2$ or $t_1, t_2 \in S_n^2$.

Without loss of generality, we can assume that $t_1, t_2 \in S_n^2$ and $s_2 \in S_n^3, s_1 \in S_n^n$. Applying Lemma 1, we can obtain that (y, t_2) is an edge in S_n^2 with $\phi(y) \in S_n^n$. Applying Lemma 2,

we can construct a Hamiltonian path $\langle y, t_1, x \xrightarrow{P(x, t_2)} t_2 \rangle$ of S_n^2 for $\phi(x) \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(s_1, \phi(y))$ of S_n^n and a Hamiltonian path $P(s_2, \phi(x))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(y))} \phi(y), y, t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, \phi(x))} \phi(x), x \xrightarrow{P(x, t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 6.

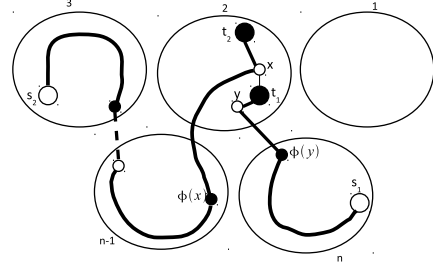


Figure 6: The illustration of case 4.1 of Lemma 6.

Case 4.2 $s_1, t_1 \in S_n^2$ or $s_2, t_2 \in S_n^2$.

Without loss of generality, we can assume that $s_2, t_2 \in S_n^2$ and $t_1 \in S_n^3, s_1 \in S_n^n$. Applying Lemma 2, we can construct a Hamiltonian path $P(s_2, t_2)$ of S_n^2 and a Hamiltonian path $P(s_1, t_1)$ in $S_n^{3, \dots, n}$. These two paths are the two spanning disjoint paths of S_n^I .

Case 4.3 $s_2, t_1 \in S_n^2$.

Suppose that $[s_2]_1 = n$ and $[t_1]_1 = 3$. Applying Lemma 1, we can obtain that (x, t_1) is an edge in S_n^2 with $\phi(x) \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, y)} y, x, t_1 \rangle$ of S_n^2 for $\phi(y) \in S_n^n$. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(y), t_2)$ of S_n^n and a Hamiltonian path $P(s_1, \phi(x))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x, t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, y)} y, \phi(y) \xrightarrow{P(\phi(y), t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 7.

Suppose that $[s_2]_1 \neq n$ or $[t_1]_1 \neq 3$. We can assume that $[t_1]_1 \neq 3$. Let (x, t_1) be an edge in S_n^2 with $\phi(x) \in S_n^3$. Applying Lemma 2, we can construct a Hamiltonian path $\langle s_2 \xrightarrow{P(s_2, y)} y, x, t_1 \rangle$ of S_n^2 for $\phi(y) \in S_n^4$. Applying Lemma 2, we can construct a Hamilto-

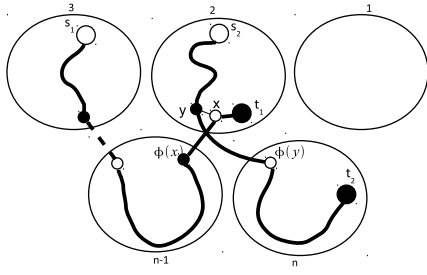


Figure 7: The illustration of case 4.3 of Lemma 6.

nian path $P(s_1, \phi(x))$ of S_n^3 and a Hamiltonian path $P(\phi(y), t_2)$ in $S_n^{4, \dots, n}$. Thus, $\langle s_1 \xrightarrow{P(s_1, \phi(x))} \phi(x), x, t_1 \rangle$ and $\langle s_2 \xrightarrow{P(s_2, y)} y, \phi(y) \xrightarrow{P(\phi(y), t_2)} t_2 \rangle$ are two spanning paths of S_n^I , as illustrated in Figure 8.

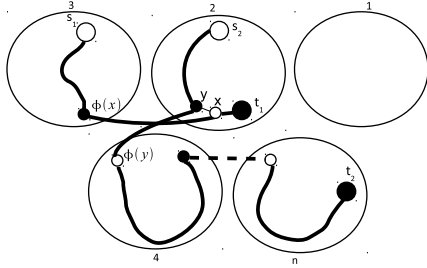


Figure 8: The illustration of case 4.3 of Lemma 6.

Case 4.4 $s_1, t_2 \in S_n^2$

Since we have not use the properties $[s_1]_1 = [t_2]_1 = 1$ in the proof of Case 4.3, The proof of this case is the same as Case 4.3 with exchanging s_1 and s_2 and exchanging t_1 and t_2 .

Case 5: The four vertices s_1, s_2, t_1, t_2 are in different subgraphs.

Without loss of generality, we can assume that $s_1 \in S_n^3$, $s_2 \in S_n^2$, $t_2 \in S_n^{n-1}$ and $t_1 \in S_n^n$. Applying Lemma 2, we can construct Hamiltonian path $P(s_1, t_1)$ of $S_n^{2, n}$. Applying Lemma 2, we can construct Hamiltonian paths $P(s_2, t_2)$ of $S_n^{3, \dots, n-1}$. Thus, $P(s_1, t_1)$ and $P(s_2, t_2)$ are two spanning paths of S_n^I . $\square$

Lemma 7 Let $I = 2, 3, \dots, n$ for $n \geq 5$. The graph $S_n^I = (V_0 \cup V_1, E)$ is hyper-Hamiltonian laceable

Proof. Let s, t, f be arbitrary vertices of S_n^I where $d(s, t)$ is even and $d(s, f)$ is odd. By the symmetry

of star graphs, we can arrange the vertices $s \in S_n^i$ and $t \in S_n^j$ for $1 \leq i < j \leq n$. Without loss of generality, we can assume that $s, t \in V_0$ and $s \in S_n^2$, $t \in S_n^n$ and $f \in V_1$. We will construct a Hamiltonian path $P(s, t)$ of $S_n^I - \{f\}$.

Case 1: $f \in S_n^i$ for some $i = 2, n$.

Without loss of generality, we can assume that $i = n$. Let y be a vertex of $(V_0 \cap S_n^n)$. Applying Lemma 2, we can construct a Hamiltonian path $P(y, t)$ of $S_n^n - \{f\}$. Applying Lemma 2, we can construct the Hamiltonian paths of S_n^i for $2 \leq i \leq n-1$. Thus, We can also construct a Hamiltonian path $P(s, \phi(y))$ of $S_n^{2, \dots, n-1}$. Thus, $\langle s \xrightarrow{P(s, \phi(y))} \phi(y), y \xrightarrow{P(y, t)} t \rangle$ is a Hamiltonian path of $S_n^I - \{f\}$.

Case 2: $f \notin (S_n^2 \cup S_n^n)$.

Without loss of generality, we can assume that $f \in S_n^3$. Let $(x, \phi(x))$ and $(y, \phi(y))$ be edges of S_n^I for $x \in (V_1 \cap S_n^1)$, $\phi(x), \phi(y) \in (V_0 \cap S_n^3)$ and $y \in (V_0 \cap S_n^4)$. Applying Lemma 2, we can construct the Hamiltonian paths $P(s, x)$ and $P(\phi(x), \phi(y))$ of S_n^2 and $S_n^3 - \{f\}$, respectively. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(y), t)$ of $S_n^{4, \dots, n-1}$. Thus, $\langle s \xrightarrow{P(s, x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(y))} \phi(y), y \xrightarrow{P(y, t)} t \rangle$ is a Hamiltonian path of $S_n^I - \{f\}$. $\square$

Theorem 1 Let u, v be two vertices of $S_n = (V_0 \cup V_1, E)$ where $d(u, v)$ is odd. Let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n where every edge of E_0 is not incident to u or v . There exists a Hamiltonian path $P(u, v)$ of $S_n - F_e$ passing through every edge of E_0 for $|F_e| \leq n-3$, $|E_0| \leq 2n-6-2|F_e|$ for $n \geq 4$.

Proof. We will prove this theorem by induction on n . This theorem is true for $n = 4$ by the brute force method. This theorem holds for $|F_e| = n-3$ by Lemma 2. In the follows, we will assume that $|F_e| \leq n-4$. By symmetry of star graph, we can arrange every faulty edge F_e in some S_n^i and at most one prescribed edge of E_0 is not in any S_n^i for $1 \leq i \leq n$. We will construct a Hamiltonian path $P(u, v)$ passing through all edges of E_0 for any $u \in V_1, v \in V_0$. Let $E_0^i = E_0 \cap E(S_n^i)$ for $1 \leq i \leq n$. Without loss of generality, we can assume that $2|F_e^1| + |E_0^1| \geq 2|F_e^k| + |E_0^k|$ for every $1 \leq k \leq n$.

Case 1: $2|F_e^1| + |E_0^1| = 2n-6$

Let $e_0 = (b, b')$ be an edge of E_0 where at

most one edge of E_0 is incident to e_0 for $b \in V_1, b' \in V_0$. And let $E_1 = E_0 - \{e_0\}$.

Case 1.1 $u, v \in S_n^1$.

Applying Lemma 5, we can construct a Hamiltonian cycle $\langle u \xrightarrow{P(u,x)} x, b \xrightarrow{P(b,v)} v, z \xrightarrow{P(z,w)} w, b' \xrightarrow{P(b',y)} y, u \rangle$ of $S_n^1 - F_e^1$ passing through every edge of E_1 with $(b, x), (b', w) \notin E_0$. Applying Lemma 6, we can construct two spanning disjoint paths $P(\phi(x), \phi(w))$ and $P(\phi(z), \phi(y))$ in $S_n^{2, \dots, n}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(w))} \phi(w), w \xrightarrow{P(w,z)} z, \phi(z) \xrightarrow{P(\phi(z), \phi(y))} \phi(y), y \xrightarrow{P(y,b')} b', b \xrightarrow{P(b,v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 1.2 One of u, v in S_n^1

Without loss of generality, we can assume that $u \in S_n^1$. Applying Lemma 5, we can construct the Hamiltonian cycle $\langle u \xrightarrow{P(u,b)} b, z \xrightarrow{P(z,b')} b', x \xrightarrow{P(x,y)} y, u \rangle$ of $S_n^1 - F_e^1$ passing through E_1 with $(b, z), (b', x) \notin E_0$.

Suppose that $v \neq \phi(x)$. Applying Lemma 6, we can construct two spanning disjoint paths $P(\phi(y), v)$ and $P(\phi(z), \phi(x))$ in $S_n^{2, \dots, n}$. Thus, $\langle u \xrightarrow{P(u,b)} b, b' \xrightarrow{P(b',z)} z, \phi(z) \xrightarrow{P(\phi(z), \phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y), v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Suppose that $v = \phi(x)$. Applying Lemma 7, we can construct the Hamiltonian path $P(\phi(z), \phi(y))$ in $S_n^{2, \dots, n} - \{v\}$. Thus, $\langle u \xrightarrow{P(u,b)} b, b' \xrightarrow{P(b',z)} z, \phi(z) \xrightarrow{P(\phi(z), \phi(y))} \phi(y), y \xrightarrow{P(y,x)} x, \phi(x) = v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 1.3 $u, v \notin S_n^1$

Applying Lemma 5, we can construct a Hamiltonian cycle $C = \langle b, x \xrightarrow{P(x,b')} b', y \xrightarrow{P(y,b)} b \rangle$ of $S_n^1 - F_e^1$ passing through every edge of E_1 with $(b, x), (b', y) \notin E_0$. Suppose that $u \neq \phi(x)$ and $v \neq \phi(y)$. Applying Lemma 6, we can construct two spanning disjoint paths $P(u, \phi(y))$ and $P(\phi(x), v)$ in $S_n^{2, \dots, n}$. Thus,

$\langle u \xrightarrow{P(u, \phi(y))} \phi(y), y \xrightarrow{P(y,b)} b, b' \xrightarrow{P(b',x)} x, \phi(x) \xrightarrow{P(\phi(x), v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Suppose that either $u = \phi(x)$ or $v = \phi(y)$. Without loss of generality, we can assume that $v = \phi(y)$. Applying Lemma 7, we can construct the Hamiltonian path $P(u, \phi(x))$ in $S_n^{2, \dots, n} - \{v\}$. Thus, $\langle u \xrightarrow{P(u, \phi(x))} \phi(x), x \xrightarrow{P(x,b')} b', b \xrightarrow{P(b,y)} y, \phi(y) = v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Suppose that both $u = \phi(x)$ and $v = \phi(y)$.

Case 1.3.1 $u, v \in S_n^i$ for some $2 \leq i \leq n$

Without loss of generality, we can assume that $u, v \in S_n^n$. We can choose an edge (z_1, z_2) of the Hamiltonian cycle C with $z_1 \in V_1$ and $\phi(z_1), \phi(z_2) \notin S_n^n$. Without loss of generality, we can denote the Hamiltonian cycle C as $\langle b, x \xrightarrow{P(x,b')} b', y \xrightarrow{P(y,z_2)} z_2, z_1 \xrightarrow{P(z_1,b)} b \rangle$. Without loss of generality, we can assume that $\phi(z_1) \in S_n^2$ and $\phi(z_2) \in S_n^{n-1}$. Let $(\phi(z_1), \phi(z_3))$ be an edge of S_n^2 with $z_3 \in S_n^n$. Applying Lemma 2, we can construct a Hamiltonian path $\langle z_3 \xrightarrow{P(z_3,a)} a, v \rangle$ of $S_n^n - \{u\}$. Suppose that $\phi(a) \in S_n^{n-1}$. Let $(\phi(z_4), \phi(z_2))$ be an edge of S_n^{n-1} with $z_4 \in S_n^2$. Let $F^{n-1}(\phi(z_4)) = \{(\phi(z_4), w) \mid [\phi(w)]_1 \neq 1 \text{ and } [\phi(w)]_1 \neq n-2\}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(a) \xrightarrow{P(\phi(a), \phi(z_5))} \phi(z_5), \phi(z_4), \phi(z_2) \rangle$ of $S_n^{n-1} - F^{n-1}(\phi(z_4))$. Thus, $z_5 \in S_n^{n-2}$. Let $F^2(\phi(z_3)) = \{(\phi(z_3), w) \mid [\phi(w)]_1 \neq 1 \text{ and } [\phi(w)]_1 \neq 3\}$. Applying Lemma 2, we can construct the Hamiltonian path $\langle \phi(z_1), \phi(z_3), \phi(z_6) \xrightarrow{P(\phi(z_6), z_4)} z_4 \rangle$ of $S_n^2 - F^2(\phi(z_3))$. Thus, $z_6 \in S_n^3$. Applying Lemma 2, we can construct the Hamiltonian path $P(z_5, z_6)$ of $S_n^{3, \dots, n-2}$. Therefore, $\langle u, x \xrightarrow{P(x,b')} b', b \xrightarrow{P(b,z_1)} z_1, \phi(z_1), \phi(z_3), z_3 \xrightarrow{P(z_3,a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(z_5))} \phi(z_5), z_5 \xrightarrow{P(z_5, z_6)} z_6 \rangle$

$z_6, \phi(z_6) \xrightarrow{P(\phi(z_6), z_4)} z_4, \phi(z_4), \phi(z_2),$
 $z_2 \xrightarrow{P(z_2, y)} y, v\rangle$ is the Hamiltonian path passing through every edge of E_0 of $S_n - F_e$.
 Suppose that $\phi(a) \in S_n^2$. Let $F^2(\phi(z_1)) = \{(\phi(z_1), w) \mid \text{for } [\phi(w)]_1 \neq 3 \text{ and } [\phi(w)]_1 \neq n\}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(a) \xrightarrow{P(\phi(a), \phi(z_4))} \phi(z_4), \phi(z_1), \phi(z_3) \rangle$ of $S_n^2 - F^2(\phi(z_1))$. Thus, $z_4 \in S_n^3$. Applying Lemma 2, we can construct the Hamiltonian path $P(z_4, \phi(z_2))$ of $S_n^{3, \dots, n-1}$. Therefore, $\langle u, x \xrightarrow{P(x, b')} b', b \xrightarrow{P(b, z_1)} z_1, \phi(z_1), \phi(z_3), z_3 \xrightarrow{P(z_3, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(z_4))} \phi(z_4), z_4 \xrightarrow{P(z_4, \phi(z_2))} \phi(z_2), z_2 \xrightarrow{P(z_2, y)} y, v \rangle$ is the Hamiltonian path passing through every edge of E_0 of $S_n - F_e$, as illustrated in Figure 9.
 Suppose that $\phi(a) \in S_n^i$ for some $3 \leq i \leq n-2$. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(z_1), \phi(z_3))$ of S_n^2 and a Hamiltonian path $P(\phi(a), \phi(z_2))$ of $S_n^{3, \dots, n-1}$. Therefore, $\langle u, x \xrightarrow{P(x, b')} b', b \xrightarrow{P(b, z_1)} z_1, \phi(z_1) \xrightarrow{P(\phi(z_1), \phi(z_3))} \phi(z_3), z_3 \xrightarrow{P(z_3, a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(z_2))} \phi(z_2), z_2 \xrightarrow{P(z_2, y)} y, v \rangle$ is the Hamiltonian path passing through every edge of E_0 of $S_n - F_e$.

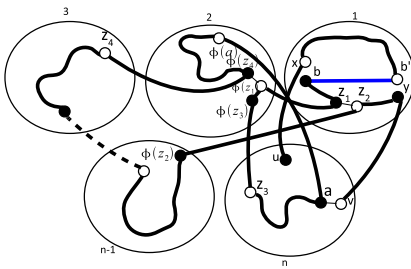


Figure 9: The Case 1.3.1 of Theorem 1.

Case 1.3.2 $u \in S_n^i$ and $v \in S_n^j$ for $2 \leq i \neq j \leq n$

Without loss of generality, we can assume that $u \in S_n^2$ and $v \in S_n^n$. We can choose an vertex $z_2 \in V_1$ of the Hamiltonian cycle C where there is not any edge of E_0 incident to z_2 and

$\phi(z_2) \in S_n^2$. Without loss of generality, we can denote the Hamiltonian cycle C as $\langle b, x \xrightarrow{P(x, b')} b', y \xrightarrow{P(y, z_2)} z_2, z_1 \xrightarrow{P(z_1, b)} b \rangle$.
 Applying Lemma 2, every S_n^i is Hamiltonian laceable and hyper-Hamiltonian laceable. Thus, we can construct the Hamiltonian path $P(\phi(z_1), \phi(z_2))$ of $S_n^{2, \dots, n} - \{u, v\}$. Therefore, $\langle u, x \xrightarrow{P(x, b')} b', b \xrightarrow{P(b, z_1)} z_1, \phi(z_1) \xrightarrow{P(\phi(z_1), \phi(z_2))} \phi(z_2), z_2 \xrightarrow{P(z_2, y)} y, v \rangle$ is the Hamiltonian path passing through every edge of E_0 of $S_n - F_e$.

Case 2: $2|F_e^1| + |E_0^1| = 2n - 7$

Let $e_0 = (b, b')$ be the edge of E_0 for $e_0 \notin S_n^1$. Suppose one of b and b' is in S_n^1 . This case is similar as Case 1. In the follows, we assume that $b, b' \notin S_n^1$.

Case 2.1 $u, v \in S_n^1$.

Applying Lemma 5, we can construct a Hamiltonian cycle $\langle u \xrightarrow{P(u, y)} y, v \xrightarrow{P(v, x)} x, u \rangle$ of $S_n^1 - F_e^1$ passing through every edge of E_0^1 . Applying Lemma 4, we can construct a Hamiltonian path $P(\phi(y), \phi(x))$ in $S_n^{2, 3, \dots, n}$ passing through e_0 . Thus, $\langle u \xrightarrow{P(u, y)} y, \phi(y) \xrightarrow{P(\phi(y), \phi(x))} \phi(x), x \xrightarrow{P(x, v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 10.

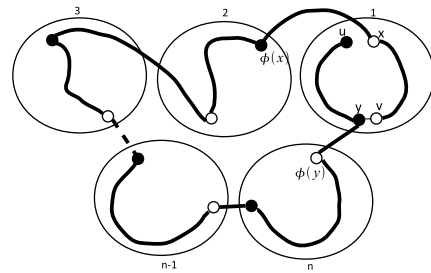


Figure 10: The Case 2.1 of Theorem 1.

Case 2.2 $u \notin S_n^1$ or $v \notin S_n^1$

Without loss of generality, we can assume that $u \in S_n^1$ or $v \in S_n^n$. Applying Lemma 5, we can construct a Hamiltonian cycle $\langle u \xrightarrow{P(u, x)} x, u \rangle$ of $S_n^1 - F_e^1$

passing through every edge of E_0^1 . Applying Lemma 4, we can construct a Hamiltonian path $P(\phi(x), v)$ in $S_n^{2,3,\dots,n}$ passing through e_0 . Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 11.

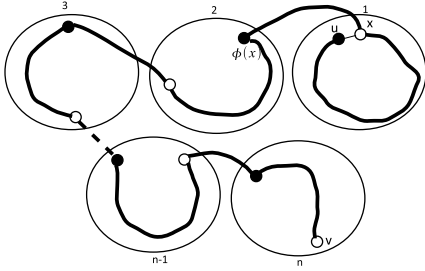


Figure 11: The Case 2.2 of Theorem 1.

Case 2.3 $u, v \in S_n^i$ for $i \neq 1$.

Without loss of generality, we can assume that $i = n$. By induction hypothesis, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u,x)} x, y \xrightarrow{P(y,v)} v \rangle$ passing through a prescribed edge of S_n^n where $[x]_1 = 1$ and there is not prescribed edge incident $\phi(x)$. By induction hypothesis, we can construct a Hamiltonian cycle $\langle \phi(x) \xrightarrow{P(\phi(x),z)} z, \phi(x) \rangle$ passing through E_0^1 of $S_n^1 - F_e^1$ with $[z]_1 \neq [y]_1$. There exists a Hamiltonian path between any two vertices with odd distance passing through a prescribed edge in S_n^j for $2 \leq j \leq n-1$. Thus, we can construct a Hamiltonian path $P(\phi(z), \phi(y))$ passing through a prescribed edge of $S_n^{2,\dots,n-1}$. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),z)} z, \phi(z) \xrightarrow{P(\phi(z),\phi(y))} \phi(y), y \xrightarrow{P(y,v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 12.

Case 2.4 $u \in S_n^i, v \in S_n^j$ for $2 \leq i < j \leq n$.

Without loss of generality, we can assume that $i = 2$ and $j = n$. Let $e_0 = (b, b')$ be the prescribed edge who is not in $E'(S_n^1)$ for $b \in V_1; b' \in V_0$.

Case 2.4.1 $b \in S_n^2$ and $b' \in S_n^n$.

Let y be a vertex of $S_n^1 \cap V_0$ where there is no prescribed edge incident to y and $[y]_1 = n$. Applying

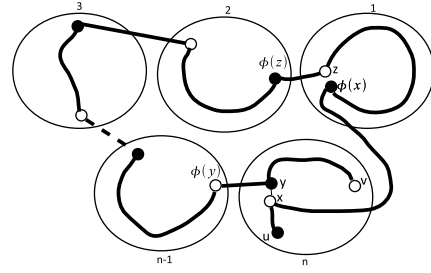


Figure 12: The Case 2.3 of Theorem 1.

Lemma 5, there exists a Hamiltonian cycle $\langle x \xrightarrow{P(x,y)} y, x \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 for $[x]_1 \neq 2$. Without loss of generality, we can assume $[x]_1 = 3$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(y) \xrightarrow{P(\phi(y),\phi(w_1))} \phi(w_1), b', \phi(w_2) \xrightarrow{P(\phi(w_2),v)} v \rangle$ of S_n^n with $w_1, w_2 \in S_n^{n-1} \cup S_n^{n-2}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u,a)} a, b, z \rangle$ of S_n^2 with $\phi(a) \in S_n^{n-2}$ and $\phi(z) \in S_n^{n-1}$.

Suppose that $w_2 \in S_n^{n-1}$. Applying Lemma 2, we can also construct a Hamiltonian path $P(\phi(a), \phi(x))$ of $S_n^{3,\dots,n-2}$ and a Hamiltonian path $P(\phi(z), w_2)$ in S_n^{n-1} . Thus, $\langle u \xrightarrow{P(u,a)} a, \phi(a) \xrightarrow{P(\phi(a),\phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y),\phi(w_1))} \phi(w_1), b', b, z, \phi(z) \xrightarrow{P(\phi(z),w_2)} w_2, \phi(w_2) \xrightarrow{P(\phi(w_2),v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 13.

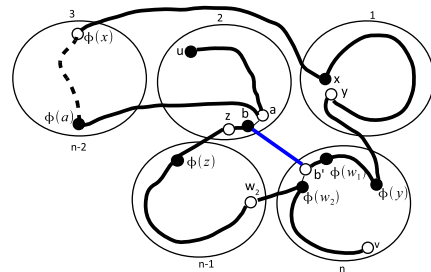


Figure 13: The Case 2.4.1 of Theorem 1.

Suppose that $w_1 \in S_n^{n-1}$. Applying

Lemma 2, we can also construct a Hamiltonian path $P(\phi(a), \phi(x))$ of $S_n^{3, \dots, n-2}$ and a Hamiltonian path $P(\phi(z), w_1)$ in S_n^{n-1} . Thus, $\langle u \xrightarrow{P(u,a)} a, \phi(a) \xrightarrow{P(\phi(a), \phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y), \phi(w_1))} \phi(w_1), w_1 \xrightarrow{P(w_1, \phi(z))} \phi(z), z, b, b' \xrightarrow{P(b',v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 14.

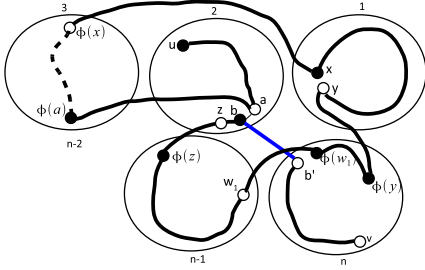


Figure 14: The Case 2.4.1 of Theorem 1.

Case 2.4.2 $b' \in S_n^2$ and $b \in S_n^n$.

Let x be a vertex of $S_n^1 \cap V_1$ where there is no prescribed edge incident to x and $[x]_1 = n$. Applying Lemma 5, there exists a Hamiltonian cycle $\langle y \xrightarrow{P(y,x)} x, y \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 for $[y]_1 \neq 2$. Without loss of generality, we can assume $[y]_1 = 3$. Applying Lemma 2, we can construct a Hamiltonian path $\langle b \xrightarrow{P(b,z_1)} z_1, \phi(x), z_2 \xrightarrow{P(z_2,v)} v \rangle$ of S_n^n for $\phi(z_1), \phi(z_2) \in S_n^{n-1} \cup S_n^{n-2}$. Suppose that $\phi(z_1) \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(u, b')$ in S_n^2 and a Hamiltonian path $P(\phi(z_1), \phi(y))$ in $S_n^{3, \dots, n-1}$. Thus, $\langle u \xrightarrow{P(u,b')} b', b \xrightarrow{P(b,z_1)} z_1, \phi(z_1) \xrightarrow{P(\phi(z_1), \phi(y))} \phi(y), y \xrightarrow{P(y,x)} x, \phi(x), z_2 \xrightarrow{P(z_2,v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 15.

Suppose that $\phi(z_2) \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(u, b')$ in S_n^2 and a Hamiltonian

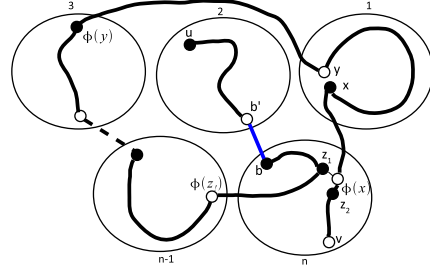


Figure 15: The Case 2.4.2 of Theorem 1.

path $P(\phi(z_2), \phi(y))$ in $S_n^{3, \dots, n-1}$.

Thus, $\langle u \xrightarrow{P(u,b')} b', b \xrightarrow{P(b,z_1)} z_1, \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y), \phi(z_2))} \phi(z_2), z_2 \xrightarrow{P(z_2,v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.4.3 $b' \in S_n^n$ and $b \in S_n^i$ for $i \neq 2, n$.

Let y be a vertex of $S_n^1 \cap V_0$ where there is no prescribed edge incident to y and $[y]_1 = n$. Applying Lemma 5, there exists a Hamiltonian cycle $\langle x \xrightarrow{P(x,y)} y, x \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 for $[x]_1 \neq n$. Suppose that $[x]_1 \neq 2$. Without loss of generality, we can assume $[x]_1 = n - 1$. Let $(\phi(a), b')$ be an edge of S_n^n for $a \in S_n^2$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(a), b', z \xrightarrow{P(z,v)} v \rangle$ of S_n^n for $\phi(z) \in S_n^{n-2}$. Applying Lemma 2, we can construct a Hamiltonian path $P(u, a)$ of S_n^2 , a Hamiltonian path $P(b, \phi(x))$ of S_n^{n-1} and a Hamiltonian path $P(\phi(y), \phi(z))$ of $S_n^{3, \dots, n-2}$. Thus, $\langle u \xrightarrow{P(u,a)} a, \phi(a), b' \xrightarrow{P(b', \phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y), \phi(z))} \phi(z), z \xrightarrow{P(z,v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 16.

Suppose that $[x]_1 = 2$. Let $(\phi(a), b')$ be an edge of S_n^n for $a \in S_n^{n-2}$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(a), b', z \xrightarrow{P(z,v)} v \rangle$ of S_n^n for $\phi(z) \in S_n^2$. Applying Lemma 2, we can construct a Hamiltonian path $\langle u \xrightarrow{P(u, \phi(z_2))}$

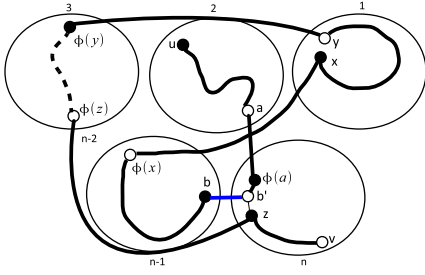


Figure 16: The Case 2.4.3 of Theorem 1.

$\phi(c_2), \phi(x), \phi(c_1) \xrightarrow{P(\phi(c_2), \phi(z))} \phi(z)$
 of S_n^2 for $c_1, c_2 \in S_n^{n-1} \cup S_n^{n-2}$.
 Suppose that $c_1 \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(b, c_1)$ of S_n^{n-1} and a Hamiltonian path $P(\phi(y), a)$ of $S_n^{3, \dots, n-2}$. Thus, $\langle u \xrightarrow{P(u, \phi(c_2))} \phi(c_2), \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), a)} a, \phi(a), b', b \xrightarrow{P(b, c_1)} c_1, \phi(c_1) \xrightarrow{P(\phi(c_1), \phi(z))} \phi(z), z \xrightarrow{P(z, v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 17.

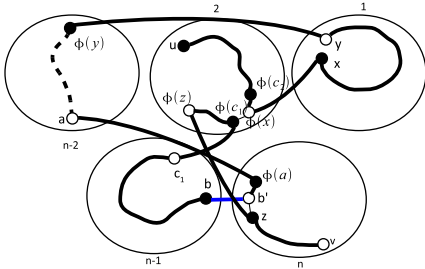


Figure 17: The Case 2.4.3 of Theorem 1.

Suppose that $c_2 \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(c_2, b)$ of S_n^{n-1} and a Hamiltonian path $P(a, \phi(y))$ of $S_n^{3, \dots, n-2}$. Thus, $\langle u \xrightarrow{P(u, \phi(c_2))} \phi(c_2), c_2 \xrightarrow{P(c_2, b)} b, b', \phi(a), a \xrightarrow{P(a, \phi(y))} \phi(y), y \xrightarrow{P(y, x)} x, \phi(x), \phi(c_1) \xrightarrow{P(\phi(c_1), \phi(z))} \phi(z), z \xrightarrow{P(z, v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.4.4 $b \notin S_n^2$ and $b' \notin S_n^n$ and

$\{[b]_n, [b']_n\} \neq \{2, n\}$.

Without loss of generality, we can assume that $b \in S_n^{n-1}$ and $b' \in S_n^{n-2}$. Let x be a vertex of $S_n^1 \cap V_1$ where there is no prescribed edge incident to x and $[x]_1 = 2$. Applying Lemma 5, there exists a Hamiltonian cycle $\langle x \xrightarrow{P(x, y)} y, x \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 for $[y]_1 \neq n-1$. Suppose $[y]_1 \neq n$. Without loss of generality, we can assume that $[y]_1 = 3$. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(y), b')$ of $S_n^{3, \dots, n-2}$ and a Hamiltonian path $P(b, v)$ of $S_n^{n-1, n}$. Thus, $\langle u \xrightarrow{P(u, \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y) \xrightarrow{P(\phi(y), b')} b', b \xrightarrow{P(b, v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 18.

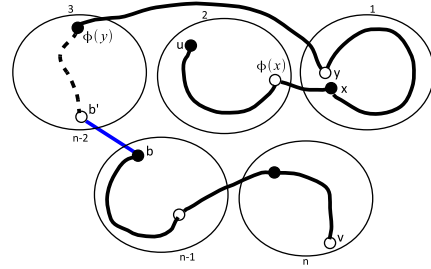


Figure 18: The Case 2.4.4 of Theorem 1.

Suppose $[y]_1 = n$. Let $(\phi(y), z)$ be an edge of S_n^n for $\phi(z) \in S_n^3$. Applying Lemma 2, we can construct a Hamiltonian path $\langle \phi(y), z, \phi(a) \xrightarrow{P(\phi(a), v)} v \rangle$ of S_n^n for $a \in S_n^{n-1}$. Applying Lemma 2, we can construct a Hamiltonian path $P(\phi(z), b')$ of $S_n^{3, \dots, n-2}$ and a Hamiltonian path $P(b, a)$ of S_n^{n-1} . Thus, $\langle u \xrightarrow{P(u, \phi(x))} \phi(x), x \xrightarrow{P(x, y)} y, \phi(y), z, \phi(z) \xrightarrow{P(\phi(z), b')} b', b \xrightarrow{P(b, a)} a, \phi(a) \xrightarrow{P(\phi(a), v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.4.5 $(b, b') \in S_n^i$ for some $i \neq 1$

Without loss of generality, we can assume that $(b, b') \notin S_n^2$. Let x be a vertex of $S_n^1 \cap V_1$ where there is no prescribed edge incident to x and $[x]_1 = 2$. Applying Lemma 5, there

exists a Hamiltonian cycle $\langle x \xrightarrow{P(x,y)} y, x \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 for $[y]_1 \neq n$. Applying Lemma 2, we can construct a Hamiltonian path $P(u, \phi(x))$ of S_n^2 and a Hamiltonian path $P(\phi(y), v)$ of $S_n^{3,\dots,n}$ passing through (b, b') . Thus, $\langle u \xrightarrow{P(u,\phi(x))} \phi(x), x \xrightarrow{P(x,y)} y, \phi(y) \xrightarrow{P(\phi(y),v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 2.4.6 $b \in S_n^1$ or $b' \in S_n^1$

Without loss of generality, we can assume that $b \in S_n^1$. Applying Lemma 5, there exists a Hamiltonian cycle $\langle b \xrightarrow{P(b,y)} y, b \rangle$ of $S_n^1 - F_e^1$ passing through E_0^1 .

Suppose that $\phi(y) \neq u$. Applying Lemma 6, we can construct two spanning disjoint paths $P(u, b')$ and $P(\phi(y), v)$ of $S_n^{2,\dots,n}$. Thus, $\langle u \xrightarrow{P(u,b')} b', b \xrightarrow{P(b,y)} y, \phi(y) \xrightarrow{P(\phi(y),v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 19.

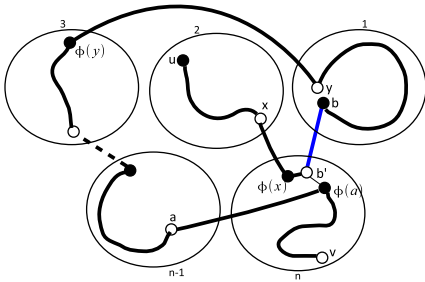


Figure 19: The illustration of case 2.4.6 of Theorem 1.

Suppose that $\phi(y) = u$. Applying Lemma 7, we can construct a Hamiltonian path $P(b', v)$ of $S_n^{2,\dots,n} - \{u\}$. Thus, $\langle u, y \xrightarrow{P(y,b)} b, b' \xrightarrow{P(b',v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 20.

Case 3: $2|F_e^k| + |E_0^k| \leq 2n - 8$ for every $1 \leq k \leq n$.

Case 3.1 $u, v \in S_n^i$ for some $1 \leq i \leq n$.

Without loss of generality, we can assume that $i = n$. By induction hypothesis, we can construct a Hamiltonian

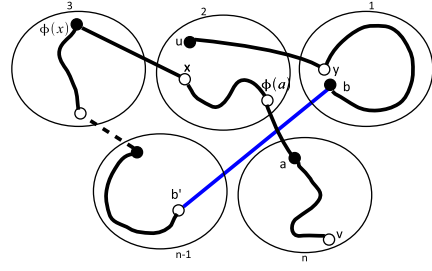


Figure 20: The Case 2.4.6 of Theorem 1.

nian path $\langle u \xrightarrow{P(u,x)} x, y \xrightarrow{P(y,v)} v \rangle$ passing through E_0^n of $S_n^n - F_e^n$ where none of E_0^j is incident to $x, y, \phi(x), \phi(y)$ for any $1 \leq j \leq n$. By induction hypothesis, there exists a Hamiltonian path between any two vertices with odd distance passing through E_0^j in $S_n^j - F_e^j$ for $1 \leq j \leq n-1$. Thus, we can construct a Hamiltonian path $P(\phi(x), \phi(y))$ passing through $E_0 - E_0^n$ of $S_n^{1,2,\dots,n-1} - (F_e - F_e^n)$. Therefore, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),\phi(y))} \phi(y), y \xrightarrow{P(y,v)} v \rangle$ is the Hamiltonian path passing through E_0 of $S_n - F_e$.

Case 3.2 $u \in S_n^i, v \in S_n^j$ for some $1 \leq i \neq j \leq n$.

Without loss of generality, we can assume that $i = 1, j = n$ and $u, b \in V_1$ and $v, b' \in V_0$.

Suppose $b \in S_n^1$ and $b' \in S_n^n$. Let (b, z) and $(b', \phi(y))$ be edges of S_n^1 and S_n^n , respectively, for $\phi(z) \in S_n^3$ and $y \in S_n^2$. By induction hypothesis, there exist the Hamiltonian paths $\langle u \xrightarrow{P(u,x)} x, b, z \rangle$ and $\langle \phi(y), b', \phi(a) \xrightarrow{P(\phi(a),v)} v \rangle$ of S_n^1 and S_n^n , respectively, for $\phi(x) \in S_n^2$ and $a \notin S_n^{n-1}$. By induction hypothesis, there exist the Hamiltonian paths $P(\phi(x), y)$ and $P(\phi(z), a)$ of S_n^2 and $S_n^{3,\dots,n-1}$, respectively. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x),y)} y, \phi(y), b', b, z, \phi(z) \xrightarrow{P(\phi(z),a)} a, \phi(a) \xrightarrow{P(\phi(a),v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 21.

Suppose that $b \in S_n^1$ or $b' \in S_n^n$ but not both. Without loss of generality, we can assume that $b \in S_n^1$ and $b' \in S_n^2$. Let $(b, \phi(z))$ be an edge of S_n^1 for $z \in S_n^{n-1}$. By induction hypothesis, there exists a

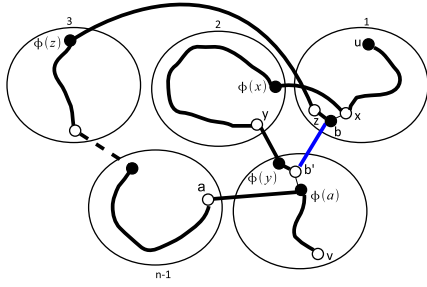


Figure 21: The Case 3.2 of Theorem 1.

Hamiltonian path $\langle u \xrightarrow{P(u,x)} x, b, \phi(z) \rangle$ of S_n^1 for $\phi(x) \in S_n^3$. By induction hypothesis, there exist the Hamiltonian paths $P(\phi(x), b')$ and $P(z, v)$ of $S_n^{2, \dots, n-2}$ and $S_n^{n-1, n}$, respectively. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x), b')} b', b, \phi(z), z \xrightarrow{P(z, v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$, as illustrated in Figure 22.

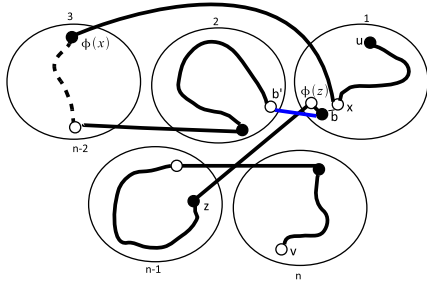


Figure 22: The Case 3.2 of Theorem 1.

Suppose $b \in S_n^n$ and $b' \in S_n^1$. By induction hypothesis, there exists a Hamiltonian path $\langle u \xrightarrow{P(u,x)} x, z \xrightarrow{P(z,b)} b \rangle$ of S_n^1 for $\phi(x), \phi(z) \notin S_n^n$. Without loss of generality, we can assume that $\phi(x) \in S_n^2$ and $\phi(z) \in S_n^3$. By induction hypothesis, there exist the Hamiltonian paths $P(b, v)$ and $P(\phi(x), \phi(z))$ of S_n^n and $S_n^{2, \dots, n-1}$, respectively. Thus, $\langle u \xrightarrow{P(u,x)} x, \phi(x) \xrightarrow{P(\phi(x), \phi(z))} \phi(z), z \xrightarrow{P(z, b')} b', b \xrightarrow{P(b, v)} v \rangle$ is a Hamiltonian path passing through E_0 of $S_n - F_e$.

Suppose that $b \notin S_n^1, b' \notin S_n^n$ and $\{b, b'\} \not\subset (S_n^1 \cup S_n^n)$. By induction hypothesis, there exists a Hamiltonian path between any two vertices with odd distance passing through E_0^j in $S_n^j - F_e^j$

for $1 \leq j \leq n$. Thus, we can construct a Hamiltonian path $P(u, v)$ passing through E_0 of $S_n - F_e$. $\square$

4 Conclusion

In this paper, we show for $n \geq 5$ let F_e be the set of faulty edges of S_n and E_0 be the edge set of some pairwise vertex-disjoint paths of S_n . All edges of E_0 lie on a Hamiltonian path $P(u, v)$ where $d(u, v)$ is odd, $|F_e| \leq n - 3$, $|E_0| \leq 2n - 6 - 2|F_e|$.

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